

MDRAI 2025-26 Final project presentations

JUN

17

Wednesday, June 17

4:00 PM - 5:30 PM GMT+2



ELISAVA ↗

Barcelona, Spain

16:30-16:55 – Session 2

- **Futures Narrative by [Kiril Traykov](#)**

How Agentic AI can strengthen Predictive AI in the apparel retail industry and improve planetary health.

The fashion industry struggles with unsold inventory



2.5 to 5.0 billion

-*Business of Fashion* estimate for the number of excess items produced by the fashion industry in 2023

21% of finished goods become unsold inventory

7% of finished + 1/3rd of returned goods are destroyed

-European Environmental Agency report on textiles in **the EU market**

Examples abound in unsold inventory mismanagement

Burberry burns bags, clothes and perfume worth millions

19 July 2018

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WHY?
LOUIS VUITTON BURNS
UNSOLD BAGS
EVERY YEAR



Why The \$4B H&M's Inventory Failure Happened

Obi Tabansi 3 August 2025 9 minutes read



- In 2018, **18%** of H&M's annual sales were classified as **unsold inventory**
- The accumulation was blamed on overproduction because of fast-changing consumer style preferences driven by social media trends
- Initiated deep markdowns, 40%+
- Created a discount outlet brand AFOUND (sold to Secret Sales in 2024)

New EU legislation to address the problem



The **Ecodesign for Sustainable Products** *framework regulation* (in force since July 2024) requires brands in the EU to report on the management of excess stock and makes it illegal to destroy unsold products (**starting 2026**). Brands need to report on:

- Quantity of unsold products
- Reasons why products became unsold
- How those products were treated (resold, recycled, donated, destroyed/landfilled)
- Measures the company is taking to reduce future disposal of unsold goods



More pressure on demand forecasting and merchandising strategy

Demand planning currently

<i>Product type</i>	<i>Demand forecast accuracy</i>
High-volume products with stable products	85-95%
Fresh products with weather-sensitive demand	70-80%
Slow-moving products with intermittent demand	50-70%

The self-prophecy trap in the data

Self-prophecy in demand planning: the universe of a brand's own consumers



Excluded group examples:



Plus-size consumer



Specialty workwear consumer



LGBT consumer

My research question

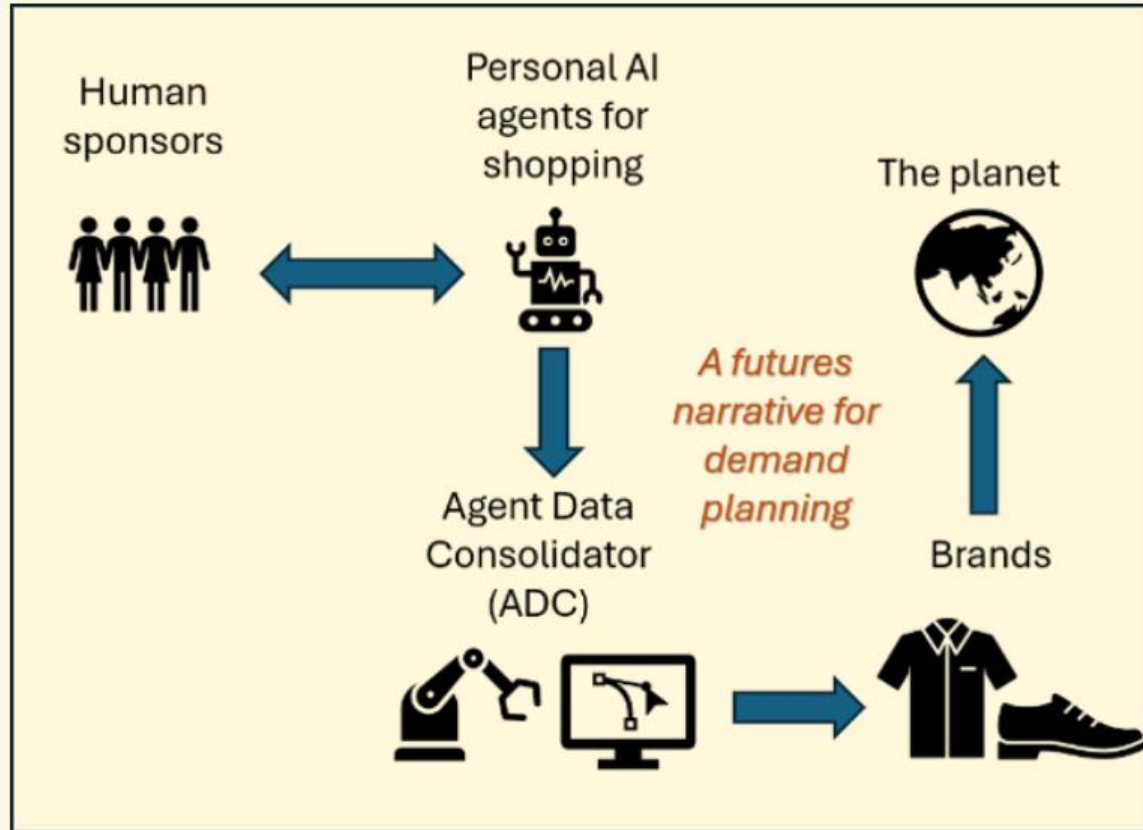
***How can agentic AI help
support EU's Ecodesign effort?***

<https://adc-narrative.netlify.app>

Anticipatory AI questions addressed in my research

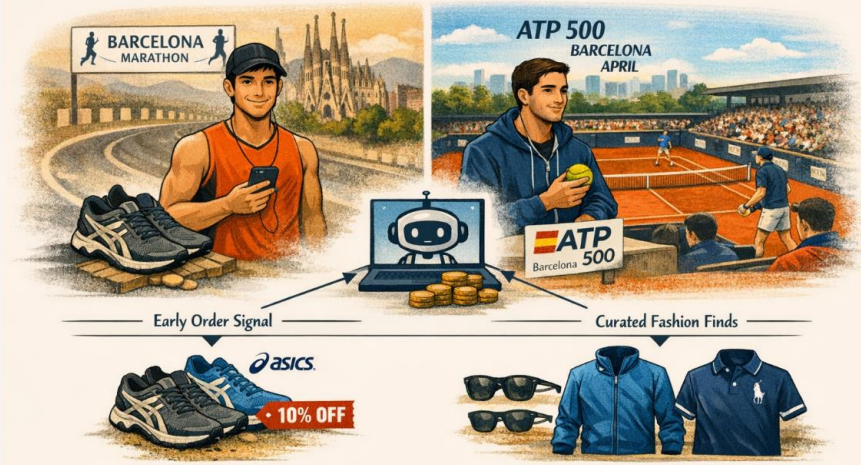
“What will happen?”	The emergence of the algorithmic consumer
“How can we use that?”	The creation of the Agentic Data Consolidator (ADC) to harness the data from the personal AI agents
“Why would we do it?”	To improve planetary health via reduction in apparel overproduction and associated environmental damages

Sociotechnical environment



Personal AI agents would increasingly handle shopping requests and become algorithmic consumers

Rafael, 26, years old in Barcelona



Maria, 40, years old in New York



Agentic Data Consolidator (ADC) prototype

AGENTIC DATA CONSOLIDATOR

ADC · Shopping Agent Demand Console

Individual agent requests below. Aggregate by any attribute. Get localized demand.

Individual Agent Entries

10 of 10 requests shown

Live feed

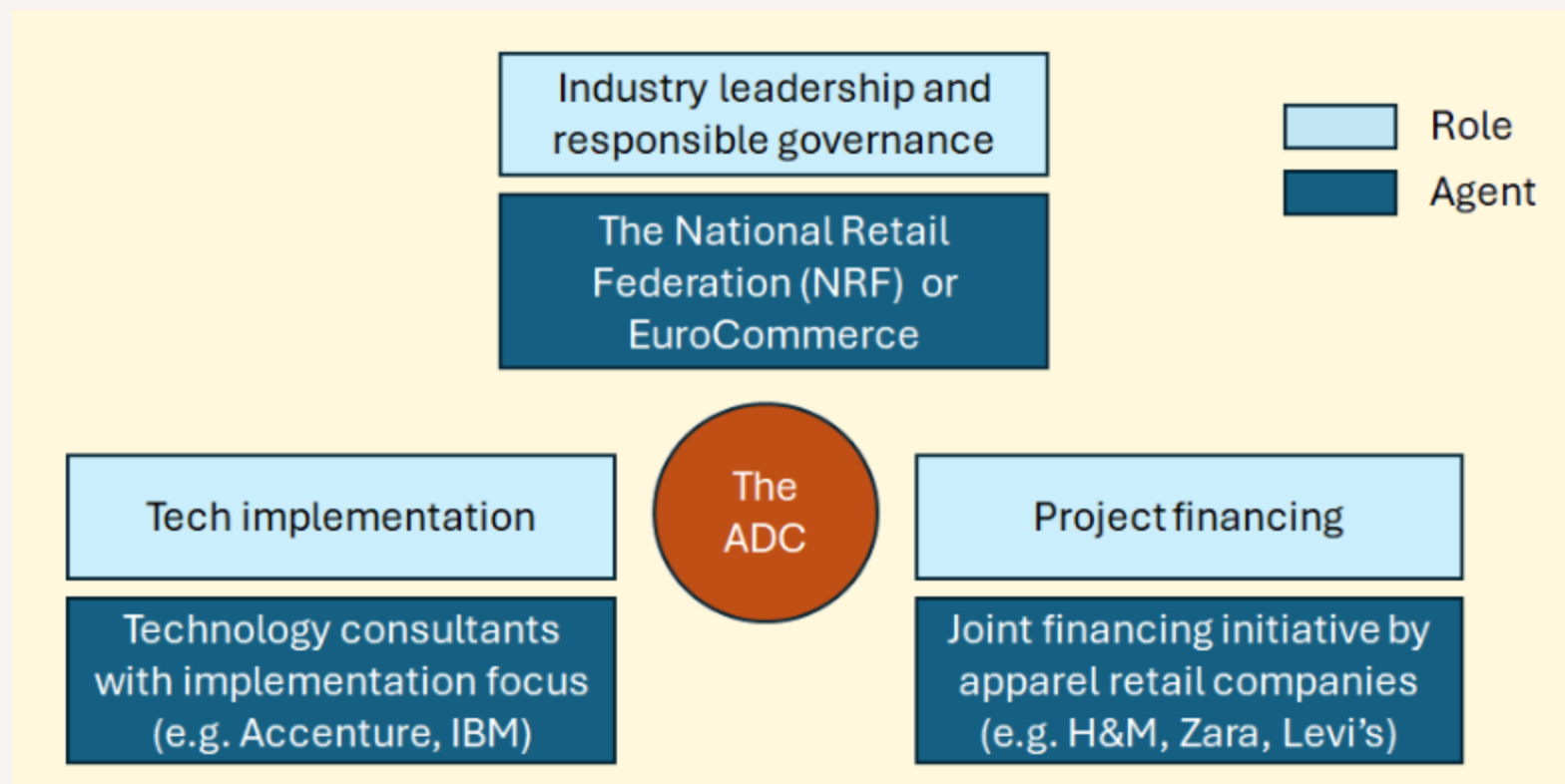
AI Agent ID	Postal Code	Product Category	Product Type	Product Style	Width	Color	Cushion	Pronation	Size	Budget	Date Needed By	Status
AA54625846	08010	Accessories	Shoes	Running	Wide	White	Regular	Neutral	13	\$80-\$100	06/31	Order
AA54625847	08010	Accessories	Shoes	Running	Standard	Black	Extra	Neutral	13	\$100-\$120	07/02	Order
AA54625848	08010	Accessories	Shoes	Running	Wide	Grey	Regular	Overpronation	13	\$120-\$140	07/05	Unmet
AA54625849	28001	Accessories	Shoes	Running	Standard	Blue	Regular	Neutral	10	\$80-\$100	06/28	Order
AA54625850	28015	Accessories	Shoes	Running	Standard	Navy	Regular	Neutral	10	\$80-\$100	07/10	Cancelled
AA54625851	41001	Accessories	Shoes	Running	Wide	Red	Regular	Neutral	9	\$80-\$100	07/12	Unmet

Timeline of 10 years

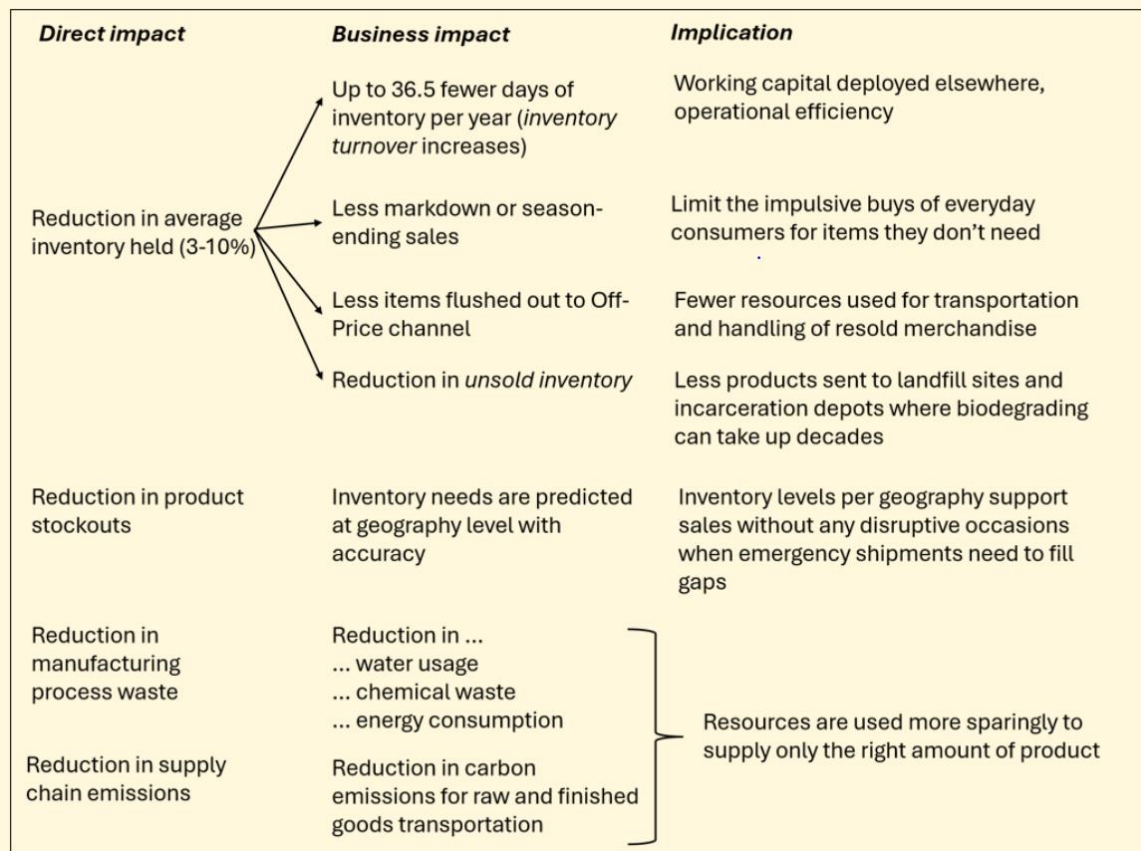
2026	2028	2030	2032	2034	2036
- Experimentation and first production deployment of AI agents, which are designed to run on PCs. - Awareness and demand for AI PCs starts growing rapidly. By year end, 54% of all manufactured PCs are AI-capable¹⁴.	10% of people (early adopters) start consistently using AI agents for regular shopping. - AI PCs start handling multi-agent workflows towards “agentic computing”.	20% of people consistently use personal AI agents for shopping. - AI PCs become the industry standard. Every laptop has a personal AI agent. - Use of traditional brand apps for shopping dramatically decreases, algorithmic consumers are in marketing focus.	35% of people consistently use personal AI agents for shopping. - The ADC is put in place by the National Retail Federation (NRF) and starts reflecting meaningful volumes of consumer product choices.	50% of people consistently use personal AI agents for shopping. - Data from the ADC is now used for demand forecasting and production planning. - First effects from the improvement in demand forecasting are reflected in apparel brands’ Sustainability Reports (as well as their Annual Financial Reports)	ADC becomes the industry standard data resource pool for demand planning

Note: In this futures narrative, the outlined developments will hold for OECD countries first

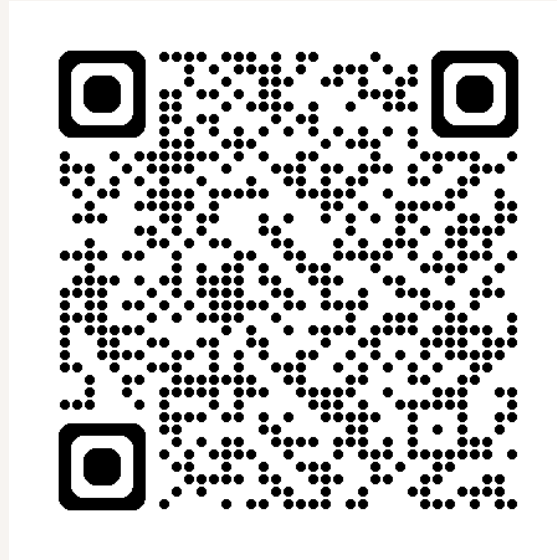
Governance model: trifecta



Planetary benefits



Learn more



<https://adc-narrative.netlify.app>

Thank you!